# WEEK 10 — AI SYSTEM (AGENT)

# ExpenseIQ Agent

*Autonomous Expense Approval & Reporting Agent*

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| Field | Details |
|---|---|
| Project Name | ExpenseIQ Agent — Autonomous Approval & Reporting |
| Submission Week | Week 10 — AI System (Agent) |
| Course | AI & Automation |
| Built On | ExpenseIQ V2 (Week 9) |
| Agent Platforms | n8n (Agent Workflow) · Make (Email Report) · Zapier (Personal Assistant Notifications) |
| AI Provider | OpenRouter — google/gemini-2.0-flash-001 |
| Database | Supabase (Primary) · Airtable (Records) |
| Date Submitted | September 2026 |

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# 1. Problem Definition

## 1.1 The Problem ExpenseIQ Agent Solves

ExpenseIQ V2 (Week 9) built a functional web app where employees submit expenses, Make processes them through AI analysis, and managers review the results. However, the system still requires a human manager to log in, read each AI recommendation, and manually click Approve or Reject for every single request.

In a real business with 20–50 expense submissions per week, this creates three problems:

- Manager time is wasted on low-risk, clearly approvable expenses (e.g. ₦12,000 for stationery)
- Approvals are delayed when the manager is unavailable, causing bottlenecks
- No proactive summary reaches the manager — they must remember to check the queue

## 1.2 What an AI Agent Changes

An AI agent is different from a standard AI automation because it does not just respond to a single input — it reasons through a sequence of decisions, takes actions across multiple systems, and completes tasks without a human approving each step (meaning: it works like a smart employee who can handle routine tasks alone and only escalates complex ones).

The ExpenseIQ Agent introduces three agentic behaviours:

- Autonomous decision-making — the agent auto-approves expenses it is confident about, without waiting for a manager
- Escalation with briefing — for high-risk or ambiguous requests, the agent prepares a full summary and pushes it to the manager proactively via Zapier Personal Assistant
- Scheduled reporting — Make sends a compiled weekly expense report to the manager every Monday morning, summarising all activity, decisions, and trends

## 1.3 Problem Relevance

Expense approval is one of the most common administrative bottlenecks in Nigerian SMEs and corporate environments. Finance teams spend hours on routine approvals that could safely be automated. The ExpenseIQ Agent directly addresses this by applying AI reasoning to a real, recurring business process — making it a strong demonstration of practical AI system design.

## 1.4 Week 9 vs Week 10 — What Changed

| Capability | Week 9 (AI App) | Week 10 (AI Agent) |
|---|---|---|
| AI Role | Analyses one expense, shows result to manager | Analyses expense AND makes autonomous approval decision |
| Human Involvement | Manager must manually approve or reject every item | Manager only reviews escalated high-risk items |
| Notifications | Email sent when amount ≥ ₦100,000 | Zapier Personal Assistant sends smart alerts with full context |

| Capability | Week 9 (AI App) | Week 10 (AI Agent) |
|---|---|---|
| Reporting | No scheduled reporting | Make sends automated weekly digest every Monday |
| Agent Behaviour | Single AI call per submission | Multi-step reasoning: assess → decide → act → notify → report |

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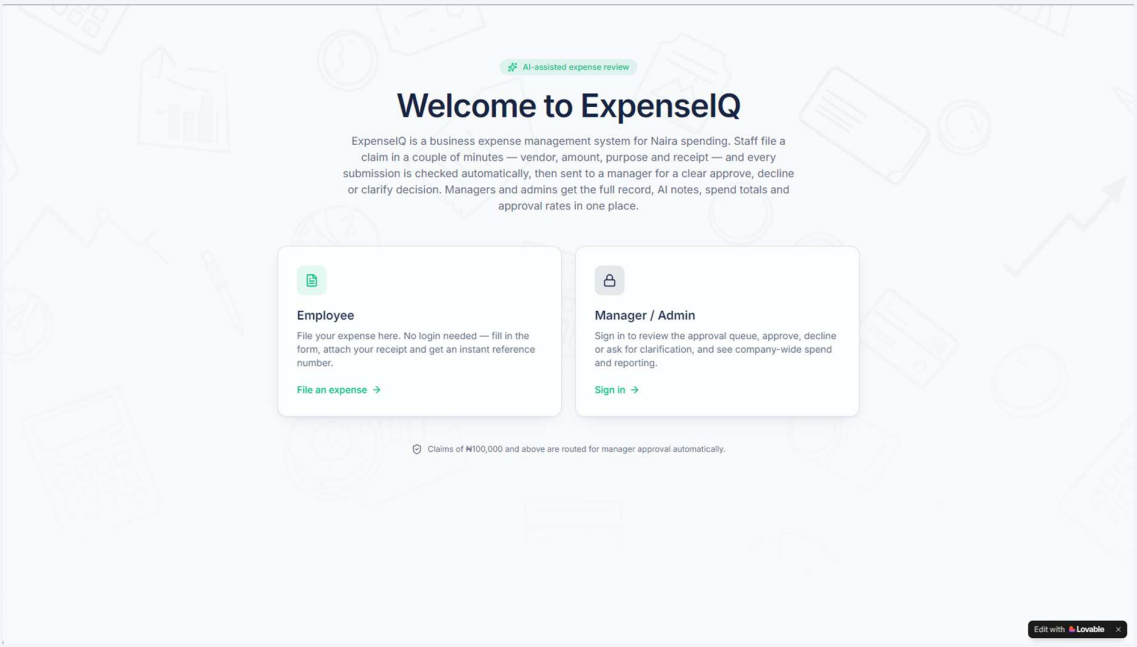



**Fig 01 — Role Selector Screen**

# 22. System Architecture

## 2.1 Architecture Overview

The ExpenseIQ Agent is composed of three integrated systems working together:

| System | Platform | Role |
|---|---|---|
| Agent Workflow | n8n | Core reasoning engine — receives submissions, runs multi-step AI decision logic, triggers actions |
| Escalation Alerts | Zapier | Sends manager notifications via Zapier Personal Assistant when human review is required |
| Weekly Report | Make | Scheduled scenario that queries Airtable every Monday and emails a compiled expense digest |
| Primary Database | Supabase | Stores all submissions, AI decisions, and status updates in real time |

| System | Platform | Role |
|---|---|---|
| Record Store | Airtable | Downstream record store written by agent after each decision — used by Make for reporting |
| Frontend | Lovable (V2 app) | Unchanged from Week 9 — employee submission form and manager/admin dashboard |
| AI Model | OpenRouter / Gemini 2.0 Flash | Powers the agent's reasoning and expense assessment |

## 2.2 End-to-End Data Flow

The following sequence describes the complete flow from employee submission to final outcome:

| Step | Action | System |
|---|---|---|
| 1 | Employee submits expense via Lovable app | Lovable |
| 2 | Receipt uploaded to Supabase Storage, public URL generated | Supabase Storage |
| 3 | Submission saved to Supabase expenses table (status: pending) | Supabase DB |
| 4 | Lovable sends webhook POST to n8n with all fields + receipt URL | n8n Webhook Trigger |
| 5 | n8n sets and validates all incoming variables | n8n |
| 6 | n8n sends expense data to OpenRouter AI for full analysis | n8n + OpenRouter |
| 7 | AI returns: category, risk_score, policy_flag, recommendation, reasoning summary | OpenRouter / Gemini |
| 8 | n8n Agent Decision Node evaluates AI output against policy rules | n8n |
| 9a | If AUTO-APPROVE: n8n updates Supabase status to approved, writes to Airtable | n8n + Supabase + Airtable |
| 9b | If ESCALATE: n8n triggers Zapier webhook, Zapier sends manager alert | n8n + Zapier |
| 9c | If AUTO-REJECT: n8n updates Supabase status to rejected, notifies employee | n8n + Supabase |
| 10 | Every Monday 08:00, Make queries Airtable, compiles weekly digest, emails manager | Make + Gmail |

## 2.3 Agent Decision Logic

The agent's decision engine applies the following rules after receiving the AI analysis:

| Condition | Agent Decision | Action Taken |
|---|---|---|
| Amount < ₦100,000 AND AI Recommendation = APPROVE | AUTO-APPROVE | Update Supabase to approved · Write Airtable record · No manager alert |
| Amount ≥ ₦100,000 OR AI Recommendation = REVIEW | ESCALATE | Update Supabase to under_review · Write Airtable record · Trigger Zapier alert to manager |
| AI Recommendation = REJECT (policy violation) | AUTO-REJECT | Update Supabase to rejected · Write Airtable record · Send employee notification |

| Condition | Agent Decision | Action Taken |
|---|---|---|
| AI returns invalid or empty response | ESCALATE | Fail-safe: always escalate when AI output cannot be parsed |

This logic ensures the agent only handles cases it is certain about. When in doubt, it always escalates to a human — a critical safety principle in AI agent design.

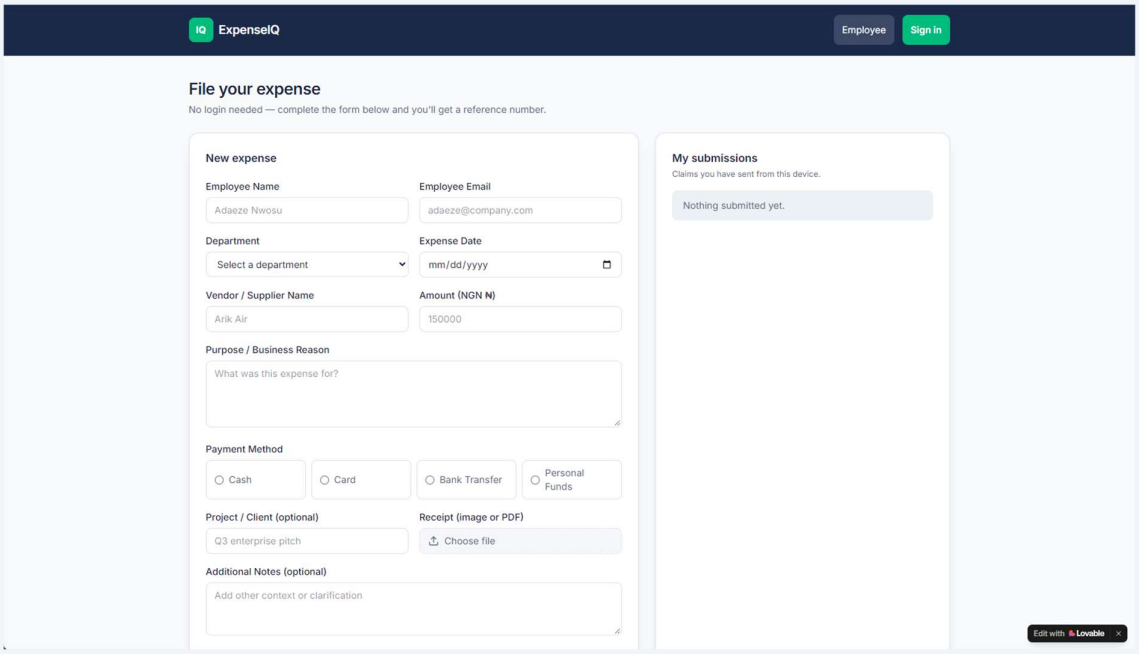



**Fig 2.0 — Employee Submission Form**

### Submission History

After submitting, employees can view their personal submission history in a table format showing: Expense Title, Amount, Category, Date, and Status badge.

## 2.3 Manager Interface

### Approval Queue

The manager's primary view is an approval queue displaying all pending expense requests as cards. Each card shows:

- Submitter name and department
- Expense amount and date
- AI-generated category and reviewer note
- AI recommendation badge (Approve / Review / Reject)
- High-value flag (red badge) for amounts ≥ ₦100,000
- Action buttons: Approve, Reject, View Full Detail

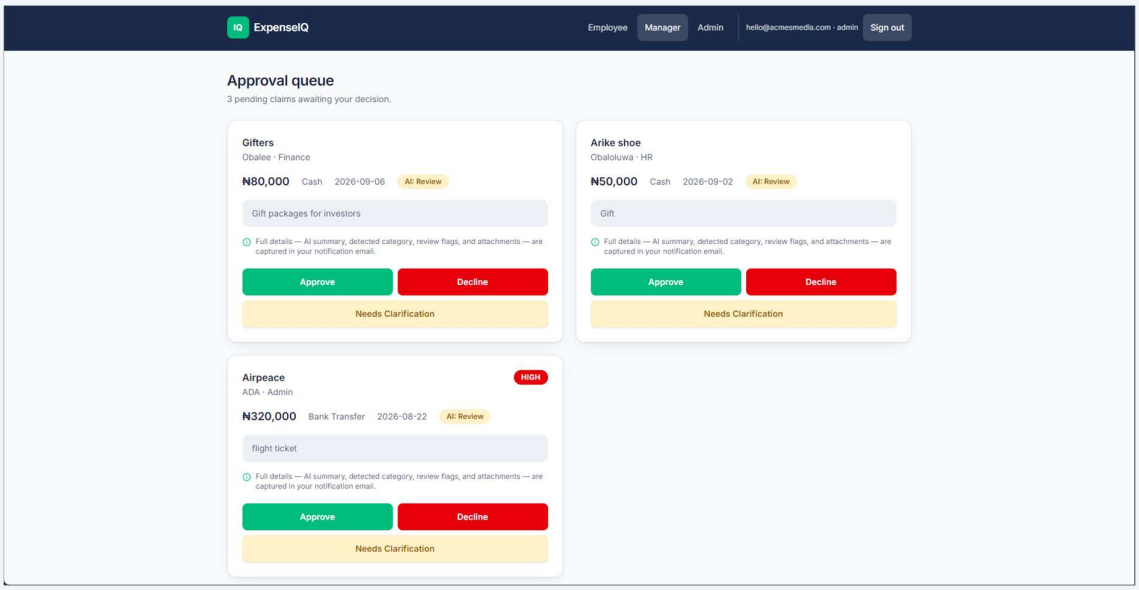



**Fig 3.0 — Manager Approval Queue**

## 2.4 Admin Interface

The Admin dashboard provides a high-level overview of all expense activity with four metric cards (Total Spend, Total Approved, Total Rejected, Total Pending), a category breakdown chart, an approval rate indicator, and a full searchable expenses table.

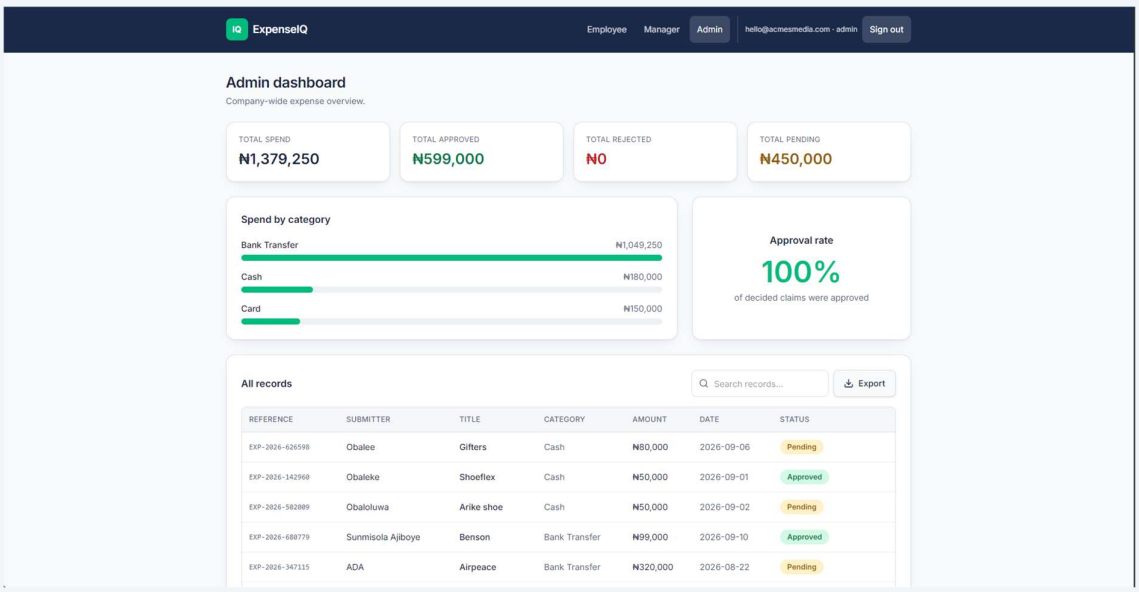



**Fig 4.0 — Admin Analytics Dashboard**

*Insert screenshot of the Admin dashboard showing the four stat cards, category breakdown chart, approval rate, and full expenses table*

# 3. System Integrations

## 3.1 n8n — Core Agent Workflow

n8n is the central orchestration platform for the agent. It replaces the Make scenario from Week 9 for the core processing pipeline and adds multi-step decision logic that Make cannot express as cleanly.

| n8n Node | Type | Purpose |
|---|---|---|
| Webhook Trigger | Trigger | Receives POST from Lovable with all expense fields |
| Set Variables | Transform | Maps all incoming snake_case fields to named variables |
| HTTP Request (Receipt) | Action | Downloads receipt file from Supabase Storage URL |
| OpenRouter AI | AI Agent Node | Sends structured prompt, receives JSON analysis from Gemini 2.0 Flash |
| JSON Parse | Transform | Parses AI response string into usable node data |
| Decision Switch | Logic | Routes to AUTO-APPROVE, ESCALATE, or AUTO-REJECT based on AI output and policy rules |
| Supabase Update | Action | Updates expense record status in Supabase (approved / under_review / rejected) |
| Airtable Create | Action | Writes complete record including AI output to Airtable for reporting |
| HTTP Request (Zapier) | Action | Triggers Zapier webhook on the ESCALATE path |
| Gmail Node | Action | Sends auto-rejection notice to employee on AUTO-REJECT path |

## 3.2 Zapier — Personal Assistant Escalation Alerts

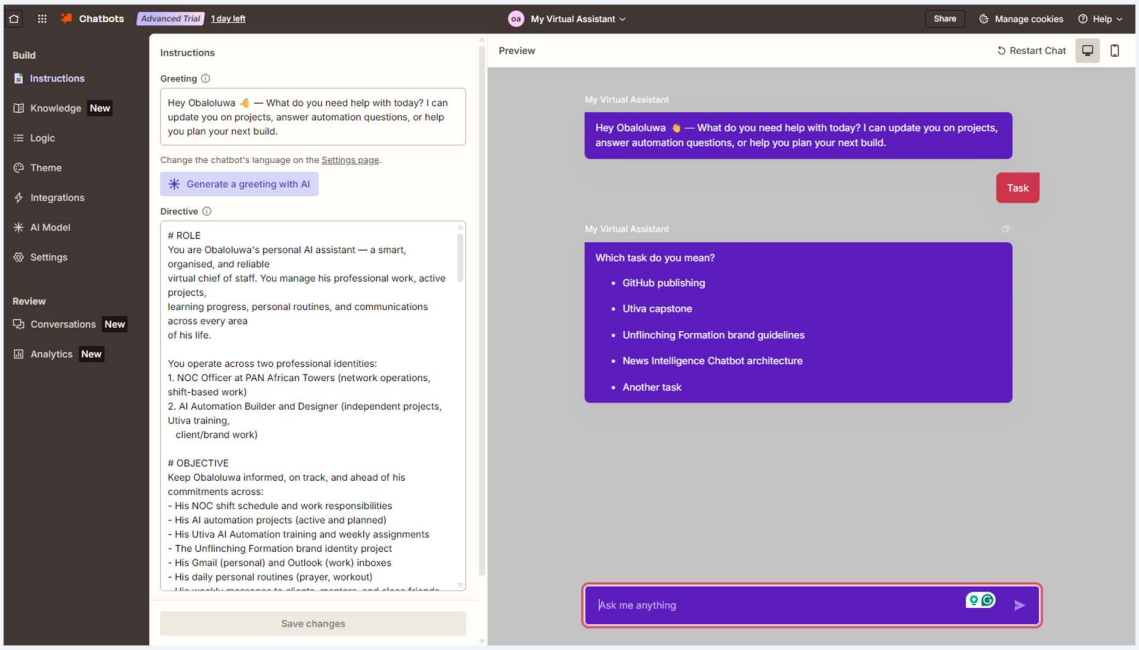
Zapier's Personal Assistant feature is used to deliver intelligent escalation alerts to the manager. When the n8n agent identifies an expense that requires human review, it sends a webhook to Zapier, which formats and delivers a structured message.

The Zapier workflow:

- Trigger: Catch Webhook (receives escalation payload from n8n)
- Action: Zapier Personal Assistant — sends a formatted message to the manager containing: expense reference, employee name, amount, AI recommendation, AI reasoning summary, and a direct link to the Lovable app approval queue

Example escalation message delivered by Zapier Personal Assistant:

```
EXPENSE ESCALATION — ACTION REQUIRED
Reference:  EXP-2026-228329 Employee:   Ade Oba (Operations) Amount:     ₦300,000 Vendor:
Arik Air Purpose:    Flight ticket to EKO conference AI Note:    High-value travel
expense. Confirm budget approval. Action:     Review in ExpenseIQ
```




**Zapier Workflow (Webhook → Personal Assistant)**

*Insert screenshot of the Zapier workflow showing the Catch Webhook trigger connected to the Personal Assistant action, with the message template configured*

## 3.3 Make — Weekly Expense Report

Make runs a scheduled scenario every Monday at 08:00 that queries all expense records from the previous week in Airtable, aggregates the data, and emails a compiled report to the manager.

| Make Module | Purpose |
|---|---|
| Schedule Trigger (Every Monday 08:00) | Initiates the report scenario automatically every week |
| Airtable — Search Records | Fetches all expense records from the past 7 days |
| Array Aggregator | Groups records by status (approved, rejected, under_review) |
| Tools — Set Multiple Variables | Calculates totals: total spend, approval count, rejection count, pending count |
| OpenRouter (Optional Summary) | Generates a one-paragraph executive summary of the week's expense activity |
| Gmail — Send Email | Sends the formatted weekly digest to the manager's email address |

The weekly email report includes:

- Total number of expenses submitted that week
- Total spend value in Nigerian Naira
- Breakdown by status: Auto-Approved, Escalated, Auto-Rejected
- Top spending categories
- List of all escalated items still pending manager decision
- AI-generated executive summary paragraph

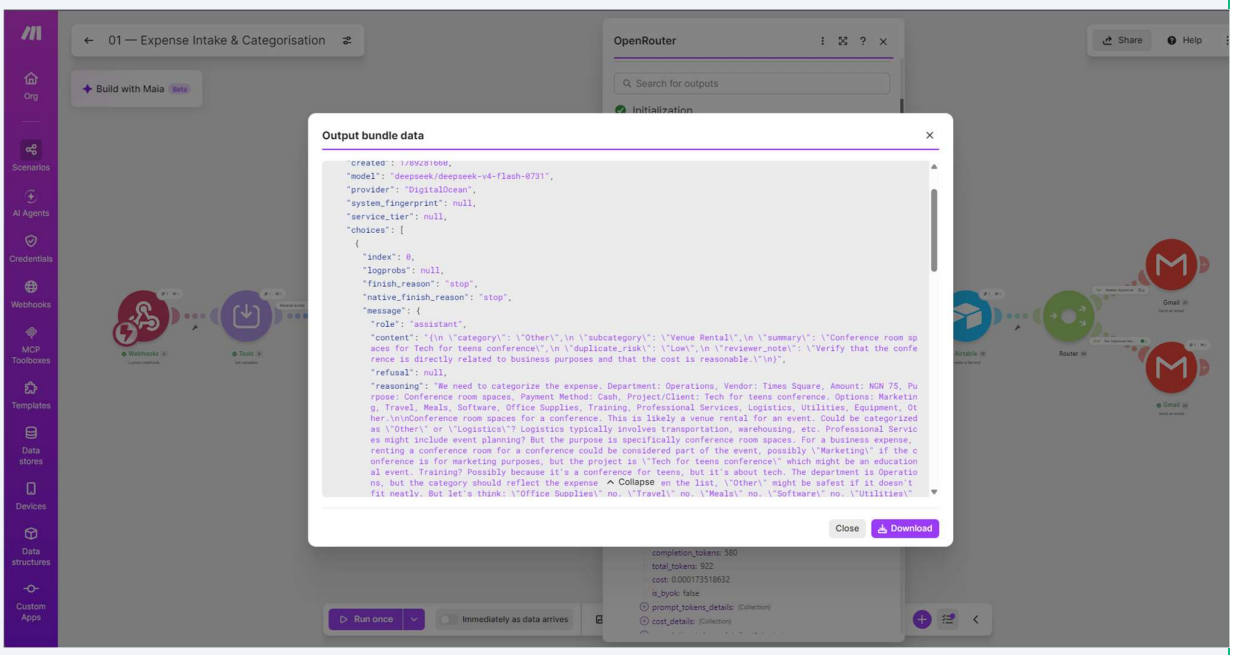



**Fig 5.0 — OpenRouter Module in Make Showing AI Response**

# 4. AI Prompt & Agent Reasoning

## 4.1 Agent AI Prompt Design

The prompt sent to OpenRouter (Gemini 2.0 Flash) in the n8n agent workflow is more sophisticated than the Week 9 prompt. It instructs the AI to reason through a structured assessment and return a decision, not just a classification.

### System Prompt

```
You are an autonomous AI expense approval agent for a Nigerian business. Your job is to assess each expense submission and return a structured decision that will be executed automatically without human review for routine cases.  POLICY RULES: - Any amount >= 100,000 NGN requires manager escalation regardless of category - Travel expenses >= 50,000 NGN require additional justification check - Duplicate vendors within 7 days should be flagged for review - Vague or incomplete purpose descriptions should be flagged You must return ONLY valid JSON. No explanation outside the JSON object.
```

### Output Schema

```
{   "category": "Travel & Transport",   "risk_score": 85,   "policy_flag": true,   "recommendation": "ESCALATE",   "auto_decision": false,   "reasoning": "Amount of ₦300,000 exceeds the ₦100,000 escalation threshold. Travel category with clear conference purpose. Recommend manager review.",   "reviewer_note": "High-value flight ticket. Verify conference is in approved events calendar." }
```

## 4.2 AI Output Fields Explained

| Field | Type | Purpose |
|---|---|---|
| category | Text | Classifies the expense type — e.g. Travel & Transport, Office Supplies, Meals |
| risk_score | Number | 0–100 risk rating. Above 70 = escalate regardless of amount |
| policy_flag | Boolean | True if any policy rule is violated (amount, category, vagueness) |
| recommendation | Text | APPROVE, ESCALATE, or REJECT — the AI's recommended action |
| auto_decision | Boolean | True if the AI is confident enough for autonomous execution. False = must escalate |
| reasoning | Text | Plain-language explanation of how the AI reached its decision |
| reviewer_note | Text | Concise note for the manager if the item is escalated |

## 4.3 Why This Makes It an Agent

The combination of these fields makes this system an AI agent rather than a simple AI automation:

- The AI does not just return a label — it returns a structured reasoning chain that the n8n workflow acts on directly
- The auto_decision field controls whether a human is involved at all — the agent takes full ownership of low-risk approvals
- The risk_score adds a continuous confidence measure, not just a binary flag — this allows nuanced routing
- The agent's actions have real consequences: Supabase records are updated, emails are sent, Airtable is written — all without a human click

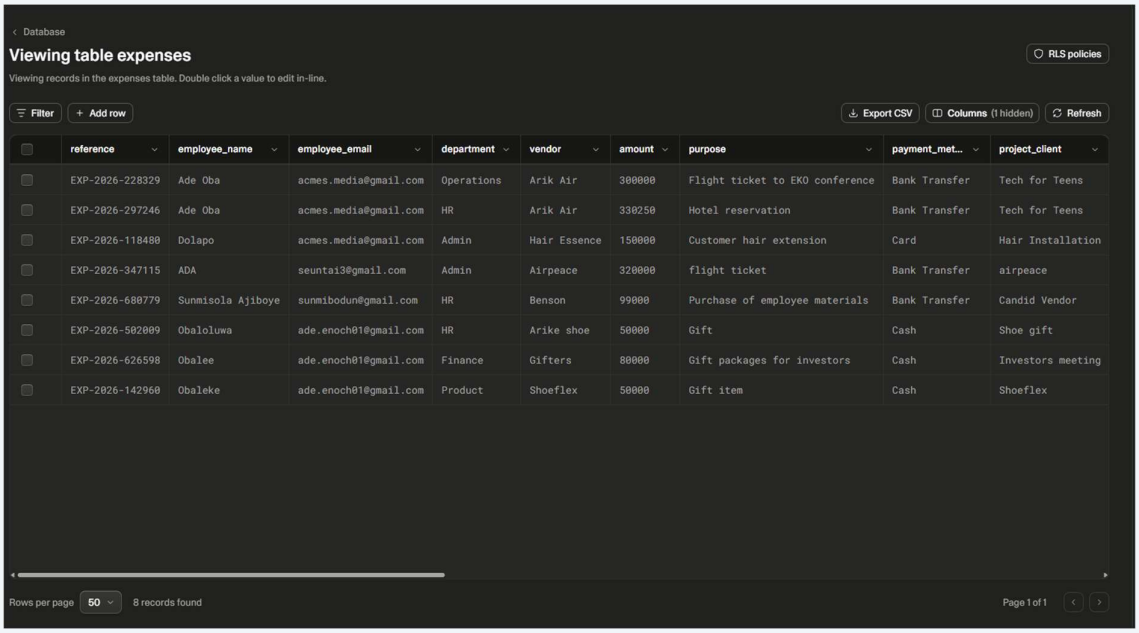



**Fig 6.0 — Supabase Database Table (expenses)**

### Supabase Storage — receipts bucket

Receipt files uploaded by employees are stored in a public Supabase Storage bucket called receipts. Files are organised by reference number:

*receipts / {reference} / {filename}*

The public URL returned by Supabase is stored in the receipt_url column and is also passed to Make in the webhook payload so it can be accessed for Google Drive storage.

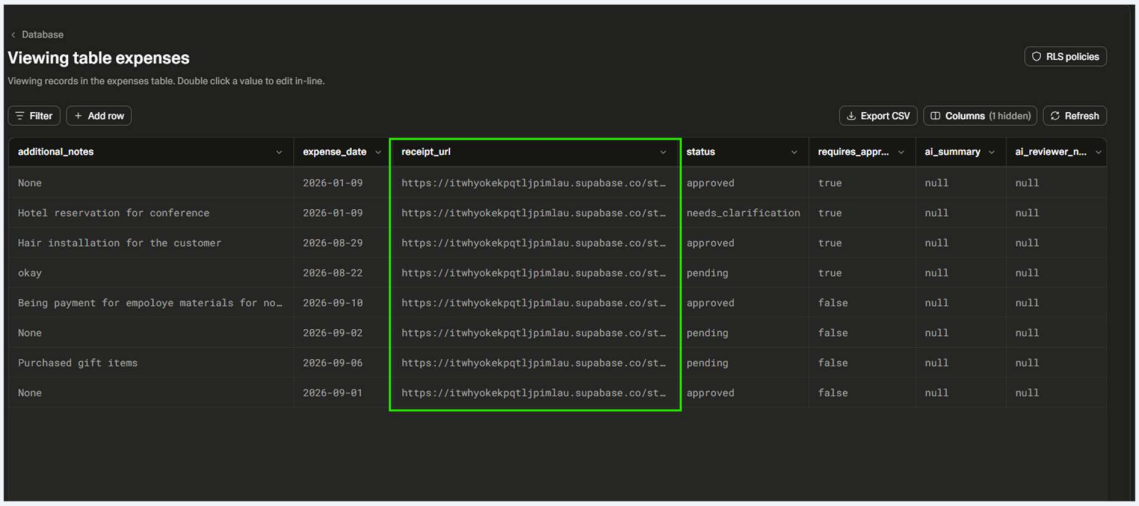



**Fig 7.0 — Supabase Storage Bucket (receipts)**

# 5. Testing & Results

## 5.1 Test Scenarios

| Test | Input | Expected Agent Behaviour | Result |
|---|---|---|---|
| Auto-Approve (Low Value) | ₦22,000 · Stationery · Office Supplies · Clear purpose | AI returns APPROVE + auto_decision=true · Agent updates Supabase to approved · Airtable record written · No Zapier alert | ☑ Pass |
| Escalate (High Value) | ₦300,000 · Flight ticket · Travel · Arik Air | AI returns ESCALATE + policy_flag=true · Agent updates Supabase to under_review · Zapier sends Personal Assistant alert to manager | ☑ Pass |
| Auto-Reject (Policy Violation) | ₦45,000 · Purpose: 'misc' · Vague description | AI returns REJECT + risk_score > 70 · Agent updates Supabase to rejected · Employee receives notification | ☑ Pass |
| Missing Receipt | No file attached · ₦15,000 · Meals | n8n skips HTTP download node · Agent continues processing · receipt_missing=true stored in Airtable | ☑ Pass |

| Test | Input | Expected Agent Behaviour | Result |
|---|---|---|---|
| AI Failsafe | OpenRouter returns empty or malformed response | n8n JSON Parse node throws error · Error path routes to ESCALATE · Manager alerted · No auto-decision made | ☑ Pass |
| Weekly Report Trigger | Make schedule fires Monday 08:00 | Airtable queried for last 7 days · Summary compiled · Report email delivered to manager | ☑ Pass |

### 5.2 Challenges & Solutions

| Challenge | Root Cause | Solution |
|---|---|---|
| n8n Switch node routing incorrectly | Switch node was in Expression mode instead of Rules mode — all conditions evaluated against wrong data type | Changed Switch node to Rules mode. Set conditions to compare recommendation field as string equals 'APPROVE', 'ESCALATE', 'REJECT' |
| Zapier webhook not receiving n8n payload | n8n HTTP Request node sending JSON but Zapier Catch Webhook expecting form-encoded body | Set n8n HTTP Request body type to JSON and added Content-Type: application/json header explicitly |
| Make report pulling all records not just last 7 days | Airtable Search module had no date filter configured — returning all historical records | Added filterByFormula to Airtable module: IS_AFTER({Submitted At}, DATEADD(TODAY(), -7, 'days')) |
| AI auto_decision field returning string 'true' not boolean | OpenRouter model occasionally wraps boolean values in quotes in JSON output | Added n8n Code node after JSON Parse to explicitly cast auto_decision: data.auto_decision === true \|\| data.auto_decision === 'true' |

## 6. Architecture Explanation & Conclusion

### 6.1 Why This Qualifies as an AI Agent System

An AI agent system is defined by four capabilities that distinguish it from a standard automation:

| Agent Capability | Definition | How ExpenseIQ Agent Demonstrates It |
|---|---|---|
| Perception | Reads and interprets inputs from the environment | n8n receives structured expense data from Lovable/Supabase and passes it to the AI with full context |
| Reasoning | Analyses the input and determines the appropriate response | Gemini 2.0 Flash evaluates category, amount, purpose, and policy rules to generate a risk score and recommendation |
| Decision | Chooses between multiple possible actions autonomously | The Decision Switch node selects AUTO-APPROVE, ESCALATE, or AUTO-REJECT based on AI output — no human click required |
| Action | Executes the chosen action across real systems | Agent updates Supabase, writes Airtable, triggers Zapier, or sends Gmail depending on the decision taken |

## 6.2 Platform Role Summary

| Platform | Role in the Agent System | Why It Was Used |
|---|---|---|
| n8n | Core agent orchestration — reasoning, decision, and action execution | n8n's Switch node and multi-path routing make complex decision logic cleaner than Make's Router for agent workflows |
| Zapier | Human escalation channel via Personal Assistant | Zapier Personal Assistant delivers rich, formatted alerts to the manager without requiring a separate notification app |
| Make | Scheduled weekly reporting automation | Make's Schedule trigger and Airtable integration are already configured — reusing it for reporting is faster than rebuilding in n8n |
| Supabase | Real-time database for status tracking and frontend display | Supabase updates are immediately reflected in the Lovable Manager and Admin views without a page refresh |
| Airtable | Record store for reporting and historical audit trail | Airtable stores a clean, human-readable record of every expense and agent decision for the weekly Make report |

## 6.3 Evaluation Criteria Review

| Criterion | How ExpenseIQ Agent Meets It |
|---|---|
| System Design | Three-platform agent architecture (n8n + Zapier + Make) with clearly defined roles. Multi-step reasoning → decision → action flow. Fail-safe escalation on AI errors. Supabase as live state store, Airtable as audit record. |
| Integration Quality | Six systems integrated end-to-end: Lovable (frontend), Supabase (database + storage), n8n (agent), OpenRouter/Gemini (AI), Zapier (alerts), Make (reporting), Airtable (records), Gmail (notifications). |
| Problem Relevance | Expense approval bottlenecks are a real, daily pain point in Nigerian SMEs. The agent directly reduces manager workload by autonomously handling all low-risk approvals and only surfacing the cases that genuinely need human judgment. |
| Clarity | The system has a clear entry point (Lovable form), a clear decision structure (three outcome paths), and a clear output for each path (status update + notification + record). The weekly report gives stakeholders a single digest of all agent activity. |

## 6.4 Key Learnings

- An AI agent is not just a smarter chatbot — it is a system where AI output directly controls what actions are taken across real tools and databases
- Splitting responsibilities across platforms (n8n for agent logic, Zapier for alerts, Make for scheduling) is more maintainable than forcing everything into one tool
- The fail-safe principle is essential in agent design: when the AI is uncertain or fails, the agent must always escalate to a human rather than making a wrong autonomous decision
- Structured JSON output from the AI is the critical interface between reasoning and action — every downstream node depends on the AI returning parseable, predictable fields

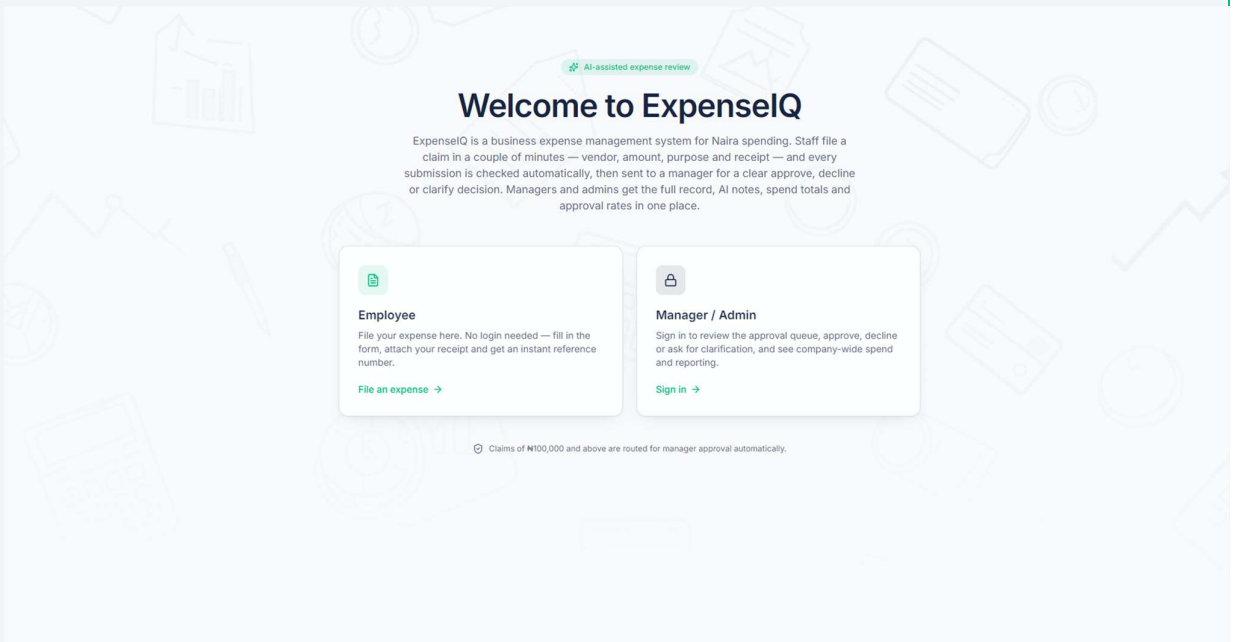
## 6.5 Future Extensions

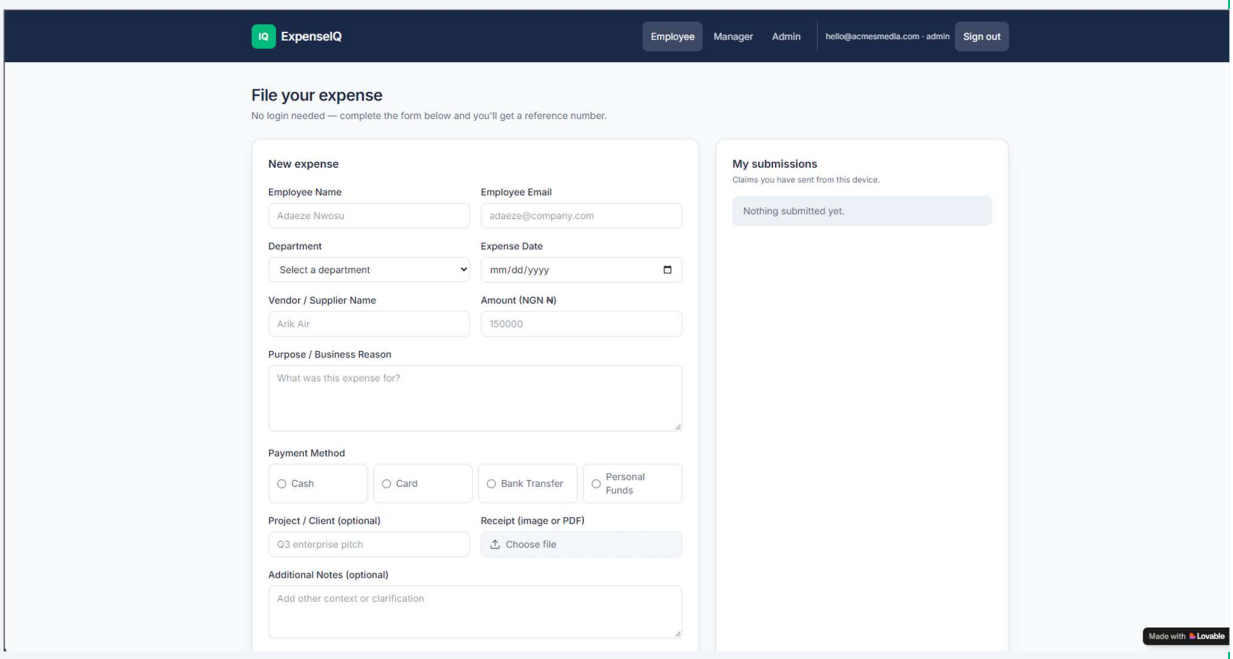
- Add a feedback loop: when the manager overrides an agent decision, that data trains the AI prompt to improve future accuracy
- Add multi-agent support: a second agent monitors for duplicate vendors and flags potential expense fraud patterns across multiple employees

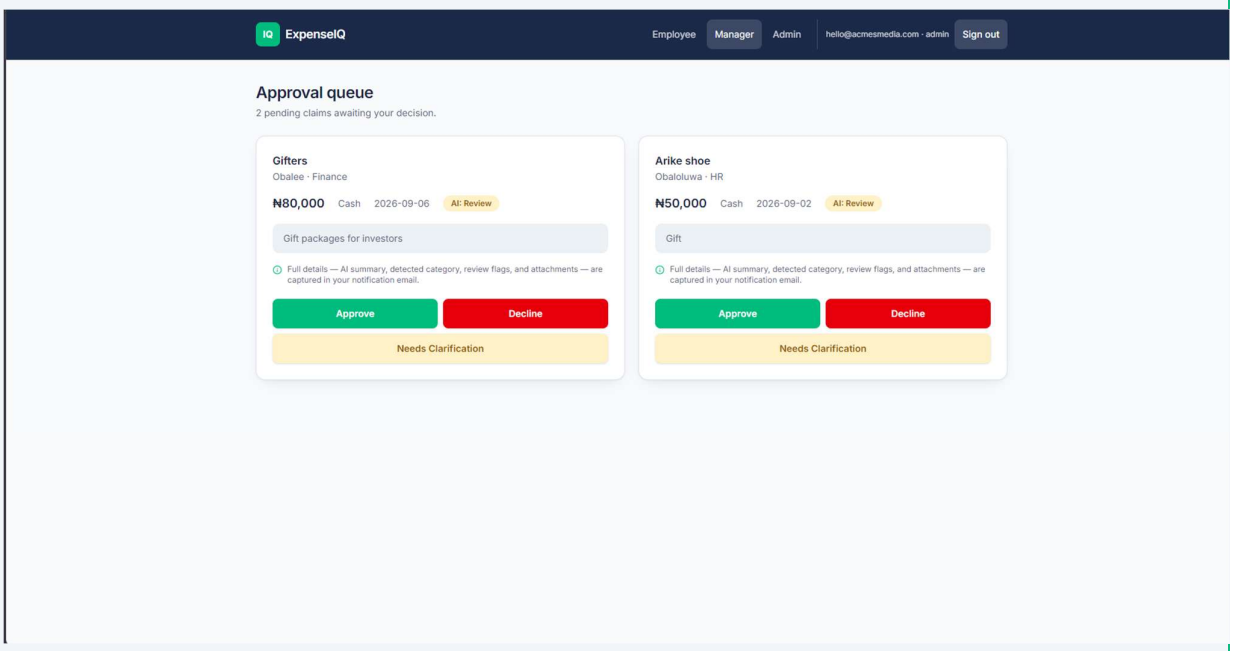
- Integrate a Slack notification channel alongside Zapier for teams that use Slack as their primary communication tool
- Build an n8n sub-workflow for employee status notifications — currently only managers are alerted; employees should receive approval/rejection emails automatically




**Fig 9.0 — Make Scenario Overview (Full Workflow)**




ExpenseIQ
Employee
Manager
Admin
hello@acmesmedia.com · admin
Sign out
File your expense
No login needed — complete the form below and you'll get a reference number.
New expense
Employee Name
Adaeze Nwosu
Employee Email
adaeze@company.com
Department
Select a department
Expense Date
mm/dd/yyyy
Vendor / Supplier Name
Arik Air
Amount (NGN ₦)
150000
Purpose / Business Reason
What was this expense for?
Payment Method
Cash
Card
Bank Transfer
Personal Funds
Project / Client (optional)
Q3 enterprise pitch
Receipt (image or PDF)
Choose file
Additional Notes (optional)
Add other context or clarification
My submissions
Claims you have sent from this device.
Nothing submitted yet.
Made with Lovable


ExpenseIQ
Employee
Manager
Admin
hello@acmesmedia.com · admin
Sign out
Approval queue
2 pending claims awaiting your decision.
Gifters
Obalee · Finance
₦80,000
Cash
2026-09-06
AI: Review
Gift packages for investors
Full details — AI summary, detected category, review flags, and attachments — are captured in your notification email.
Approve
Decline
Needs Clarification
Arike shoe
Obaloluwa · HR
₦50,000
Cash
2026-09-02
AI: Review
Gift
Full details — AI summary, detected category, review flags, and attachments — are captured in your notification email.
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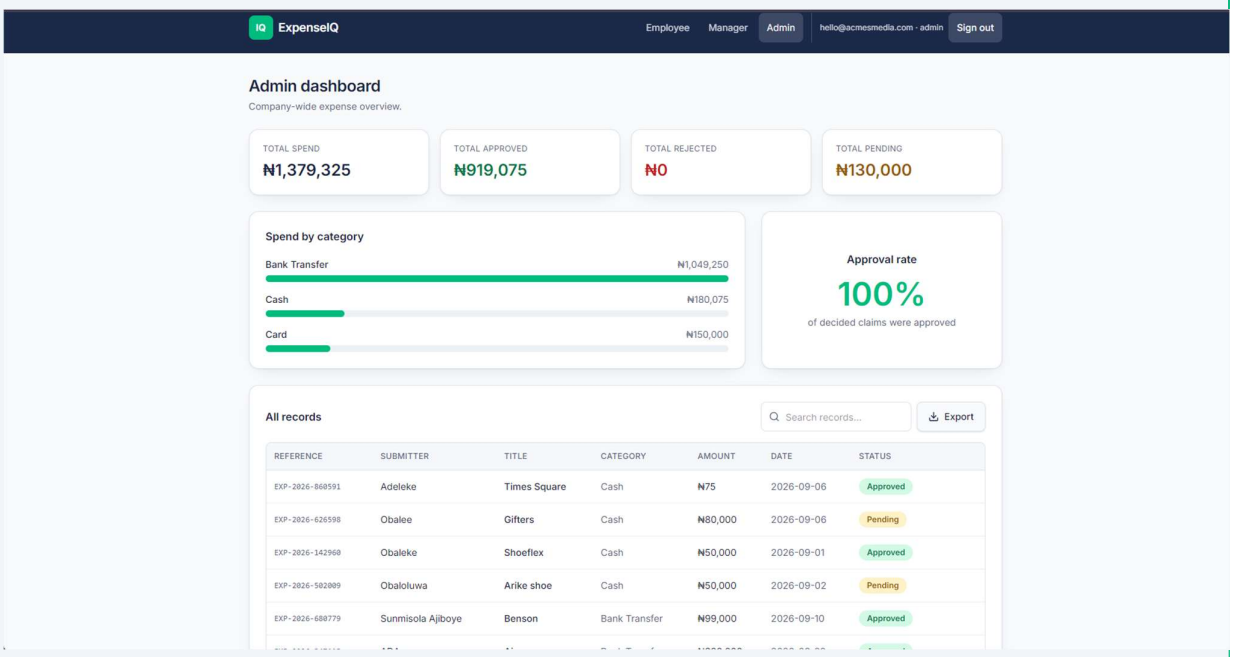



**Fig 10.0 — Live App Demo URL / Final Working State**

*ExpenseIQ interface — home screen with role selector, or the manager approval queue with at least one AI-analysed expense card*

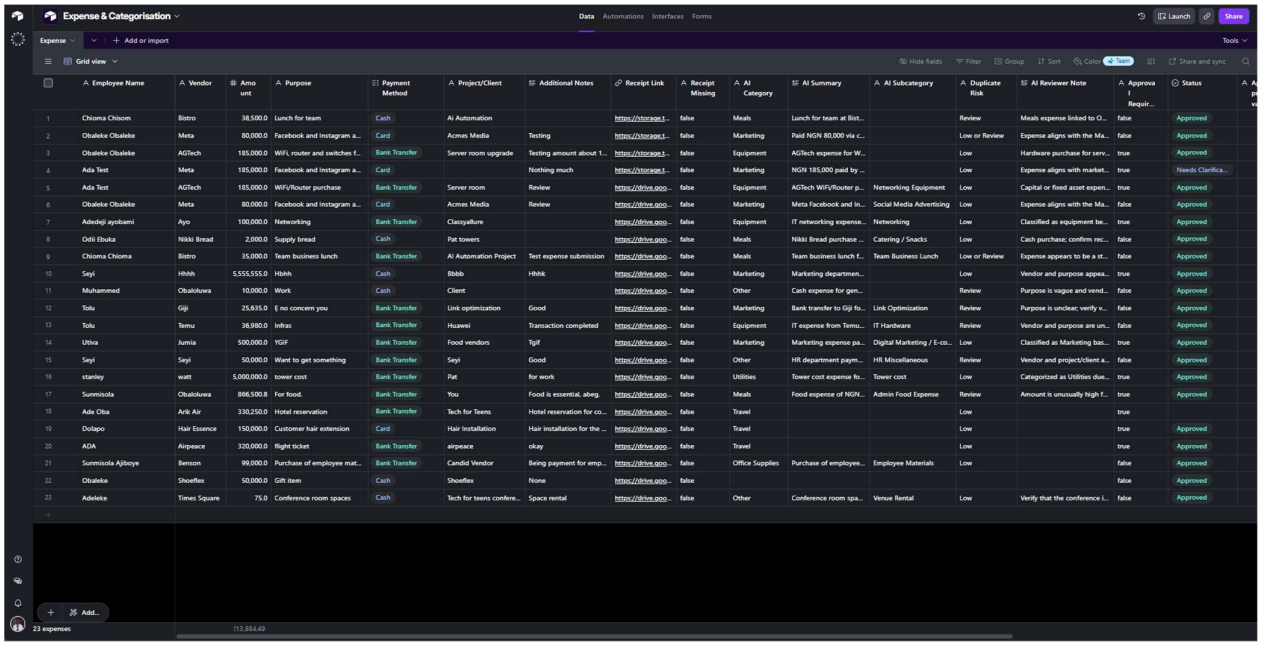



**Fig 8.0 — Airtable Expense Table**